

Assembly changes what Vaxrank can claim

Actual RNA reconstruction, translation, epitope prediction, and vaccine-selection outcomes

A DNA event is not an antigen. Vaxrank requires a defensible mutant protein before ranking epitopes.		
VARIANT	RNA / TRANSLATION EVIDENCE	VAXRANK OUTCOME
PIP5K1A p.G474fs Assembly-dependent frameshift indel	ONT assembly: 3 ALT, 0 OTHER; 49-aa mutant protein	SELECTED
H1-2 repetitive 15-nt deletion Large in-frame deletion	ONT assembly: 13 ALT, 2 OTHER; no passing target binder	NO TARGET BINDER
ATP5MG::KMT2A Structural variant / gene fusion	One coding hypothesis; five donor-CDS-unavailable alternatives	HELD OUT
TPST1::CRCP T2 Structural variant / gene fusion	22 long reads rescue expression; CDS/frame remains unresolved	NO TRANSLATION
PREDICTION CONTEXT NetMHCpan 4.2c; patient class-I alleles: HLA-A*01:01, HLA-B*08:01, HLA-B*27:05, HLA-C*01:02, HLA-C*07:01		

PIP5K1A p.G474fs

chr1:g.151242179delG (PIP5K1A p.G474fs) | Osteosarcoma T1 | Assembly-dependent frameshift indel

SELECTED

Long-read RNA assembly establishes an expressed frameshift protein and Vaxrank selects a supported long peptide.

1 DNA-ONLY INTERPRETATION

The genomic deletion predicts a frameshift, but DNA alone cannot establish the expressed haplotype, mutant protein extent, or junction-spanning peptide sequence.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Oxford Nanopore long-read RNA | T1 ONT RNA window

ALT 3 | REF 36 | OTHER 0 | top protein 2

Vaxrank RNA gates: **PASS**

3 ASSEMBLED OR TRANSLATED PROTEIN

VFKKIPLKPSPSKKFRSGSSFRRRAAPVATPALLTSHRSLGNTRHK

4 VACCINE-CONSTRUCT DECISION

SSFRRRAAPVATPALLTSHRSLGNT

Combined score 14.674 | target epitope score 8.472

This is the positive assembly-dependent result: RNA reconstruction changes an abstract frameshift call into a ranked vaccine candidate with multiple class-I binders.

Selection is computational and is not a clinical recommendation.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
RRAAPVATPAL	HLA-C*01:02	116.92	0.352	1.254
AAPVATPAL	HLA-C*01:02	3.09	0.007	0.998
AAPVATPALL	HLA-C*01:02	18.97	0.080	0.988
RAAPVATPAL	HLA-C*01:02	19.49	0.082	0.988
APVATPAL	HLA-C*01:02	77.44	0.254	0.944
LLTSHRSL	HLA-B*08:01	148.99	0.182	0.875

H1-2 repetitive 15-nt deletion

chr6:g.26055825_26055839delCCTTGGGCTTAGCGG | Osteosarcoma T1 | Large in-frame deletion

NO TARGET BINDER

RNA assembly succeeds and passes Vaxrank's evidence gates, but no target epitope passes the default prediction filters.

1 DNA-ONLY INTERPRETATION

A 15-nt deletion inside a repetitive histone sequence does not by itself identify the expressed protein context or a useful class-I target.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Oxford Nanopore long-read RNA | T1 ONT RNA window

ALT 13 | REF 33 | OTHER 2 | top protein 11

Vaxrank RNA gates: **PASS**

3 ASSEMBLED OR TRANSLATED PROTEIN

VAKPKKAAKSAAKAVKPKVVKPKKAAPKKK

4 VACCINE-CONSTRUCT DECISION

No peptide selected for a vaccine construct.

Assembly is necessary but not sufficient. The expressed sequence is resolved, yet Vaxrank correctly returns no vaccine candidate for the configured patient HLA set.

A zero-target result is retained rather than hidden or forced into selection.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
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No target epitope passed the configured filters.

ATP5MG::KMT2A

sample-specific fusion junction | Osteosarcoma T0 | Structural variant / gene fusion

HELD OUT

A coding fusion hypothesis exists, but competing noncoding donor hypotheses make the translated product ambiguous; Vaxrank holds it out.

1 DNA-ONLY INTERPRETATION

The DNA rearrangement supports a junction but cannot choose the sample-specific transcript, CDS start, reading frame, or translated junction product.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Illumina short-read RNA assembly | T0 Personalis junction window

Junction assembly evaluated; allele-fragment counts are not available

Vaxrank RNA gates: **HELD OUT BEFORE RANKING**

3 ASSEMBLED OR TRANSLATED PROTEIN

MAQFVRNLVEKTPALVNG

4 VACCINE-CONSTRUCT DECISION

No peptide selected for a vaccine construct.

The adapter preserves the plausible 18-aa translation for audit while preventing an ambiguous fusion from entering vaccine ranking.

Default NetMHCpan filtering also produced no targetable epitope for the coding hypothesis.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
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No target epitope passed the configured filters.

TPST1::CRCP T2

DNA-positive, matched short-read-call-negative junction | Osteosarcoma T2 | Structural variant / gene fusion

NO TRANSLATION

DNA and 22 long reads support the junction, but the donor joins before its annotated CDS; no defensible fusion protein exists.

1 DNA-ONLY INTERPRETATION

The rearrangement is real enough to investigate, but a breakpoint is not a coding transcript and does not define a junction-spanning antigen.

2 SAMPLE-SPECIFIC RNA EVIDENCE

Oxford Nanopore long-read RNA | T2 CTAT-LR-Fusion

22 junction-spanning long reads; no translated protein

Vaxrank RNA gates: **NOT APPLICABLE: NO PROTEIN**

3 ASSEMBLED OR TRANSLATED PROTEIN

Not established

4 VACCINE-CONSTRUCT DECISION

No peptide selected for a vaccine construct.

This is a genuine long-read rescue of expression, not a Vaxrank antigen.
The protein and peptide claims remain withheld until the coding frame is established.

No translated sequence means no epitope prediction or vaccine ranking is scientifically valid.

5 TARGET EPITOPES PASSING DEFAULT FILTERS

PEPTIDE	ALLELE	IC50 NM	RANK %	SCORE
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Not predicted because no translated sequence exists.

What is measured, inferred, and withheld

1 RECONSTRUCT

Isovar groups RNA fragments by allele, assembles supported sequence, and translates only when the coding frame is defensible.

EVIDENCE RETAINED

ALT, REF, and OTHER fragment counts; supporting-fragment count; platform; sample; transcript or junction provenance.

2 PREDICT AND RANK

Vaxrank enumerates mutant-protein windows, predicts class-I binding through mhctools and Topiary, and applies the configured RNA and epitope gates.

ALLELE POLICY

The clinical null allele HLA-A*01:11N is not assessed. The five expressed class-I alleles are evaluated.

3 WITHHOLD WHEN NEEDED

Junction support alone does not establish a CDS, reading frame, translated product, or junction-spanning peptide. Ambiguous and unresolved fusion hypotheses remain outside vaccine selection.

NO EMPTY-CATEGORY FUDGING

Assembly success may still produce no target binder. That is a negative Vaxrank result, not a missing result.

SOFTWARE AND MODEL

isovar 1.17.0 | mhctools 3.44.0 | netMHCpan 4.2c | topiary 5.55.1 | vaxrank 3.18.2

<https://osteosarc.com/variant/PIP5K1A-chr1-151242178/>
https://osteosarc.com/variant/PIP5K1A-chr1-151242178/selection_validation/validated.json
<https://osteosarc.com/>
<https://osteosarc.com/fusions/>
[tests/data/isovar_fusions/README.md](https://osteosarc.com/fusions/README.md)
https://osteosarc.com/ctat_lr_fusion/TPST1--CRCP/